Roll No. ___________________

# KIET Group of Institutions

## CT Examination (2024-2025) EVEN Semester

**Department: CSE/IT/CS/CSIT/CSEAI/CSEAIML**
**Course: B.Tech**
**Year: III**
**Semester: VI**
**Subject Name: Computer Networks**
**Subject Code: BCS 603**
**Duration: 2Hrs**
**Max. Marks: 60**
**Note: Attempt all the questions of each section**

| Q.No. | | Question | CO | BL/KC* |
|---|---|---|---|---|
| **Section-A** | | | **(10X2=20)** | |
| 1 | **a.** | Explain the term Computer Networks along with its main purpose. | CO1 | 2F/C |
| | **b.** | Discuss the advantage of switching technique. | CO1 | 2F/C |
| | **c.** | Explain the main need of TCP/IP Model. | CO1 | 2F/C |
| | **d.** | Name the two network devices that operate at the Physical Layer. | CO1 | 1F/C |
| | **e.** | Differentiate between guided and unguided transmission media. | CO1 | 2/F |
| | **f.** | Describe the term “Packetizing” in reference of data link layer. | CO2 | 2F/C |
| | **g.** | Differentiate between Byte Stuffing and Bit Stuffing. | CO2 | 2F/C |
| | **h.** | Distinguish Single Bit Error and Burst Error. | CO2 | 2F/C |
| | **i.** | Differentiate between Switch and Router. | CO2 | 2F/C |
| | **j.** | Describe Internet Service Provider (ISP) in connecting users to the Internet. | CO2 | 1F/C |
| **Section-B** | | | **(5X4=20)** | |
| **Q.No.** | | **Question** | **CO** | **BL/KC*** |
| 2 | **a.** | Discuss the LAN, WAN, Intranet, Extranet and VPN. | CO1 | 2F/C |
| | | OR | | |
| | **b.** | Discuss the switching techniques with its types. | | |
| 3 | **a.** | Illustrate the Frame. Discuss the variable size framing mechanism. | CO2 | 3F/C |
| | | OR | | |
| | **b.** | If the data **11010011101100** is to be transmitted using the generator polynomial $\mathbf{x^3 + x + 1}$, determine the CRC code. | | |
| 4 | **a.** | Define signal encoding and its classification used in the Physical Layer. Write 1101001 in all signal encoding schemes | CO1 | 1F/C |
| | | OR | | |
| | **b.** | State the major types of transmission Impairment that impact network performance with the help of suitable diagram. | | |
| 5 | **a.** | A receiver receives the 8-bit code 11010111, which was sent using odd parity. Does the received data have an error? Use parity checking mechanism to detect. | CO2 | 3F/C |
| | | OR | | |
| | **b.** | A sender wants to send data of five character: A B ESC FLAG ESC in a frame. Determine the created frame using byte stuffing mechanism. | | |
| **Section-C** | | | **(2X10=20)** | |
| **Q.No.** | | **Question** | **CO** | **BL/KC*** |
| 6 | **a.** | Determine a network topology for a small office and justify your designed network, used transmission media, and interconnecting devices. | CO1 | 3F/C |
| | | OR | | |
| | **b.** | Illustrate the OSI Reference Model, including the role of each layer and key protocols. | | |
| 7 | **a.** | Discuss the key functional properties of Data Link Layer in TCP/IP Model. | CO2 | 2F/C |
| | | OR | | |
| | **b.** | Discuss the different error detection techniques used in the Data Link Layer, with examples and their effectiveness. | | |

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- **Bloom’sLevel(BL)-Bloom’s taxonomy framework is planning and designing of assessment of student’s learning.**
- *Knowledge Categories (KCs): **F**-Factual, **C**-Conceptual, **P**-Procedural, **M**-Metacognitive
- #Reference to Competitive Exams(GATE, GPAT, CAT, GRE, TOFEL NET,etc.)

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